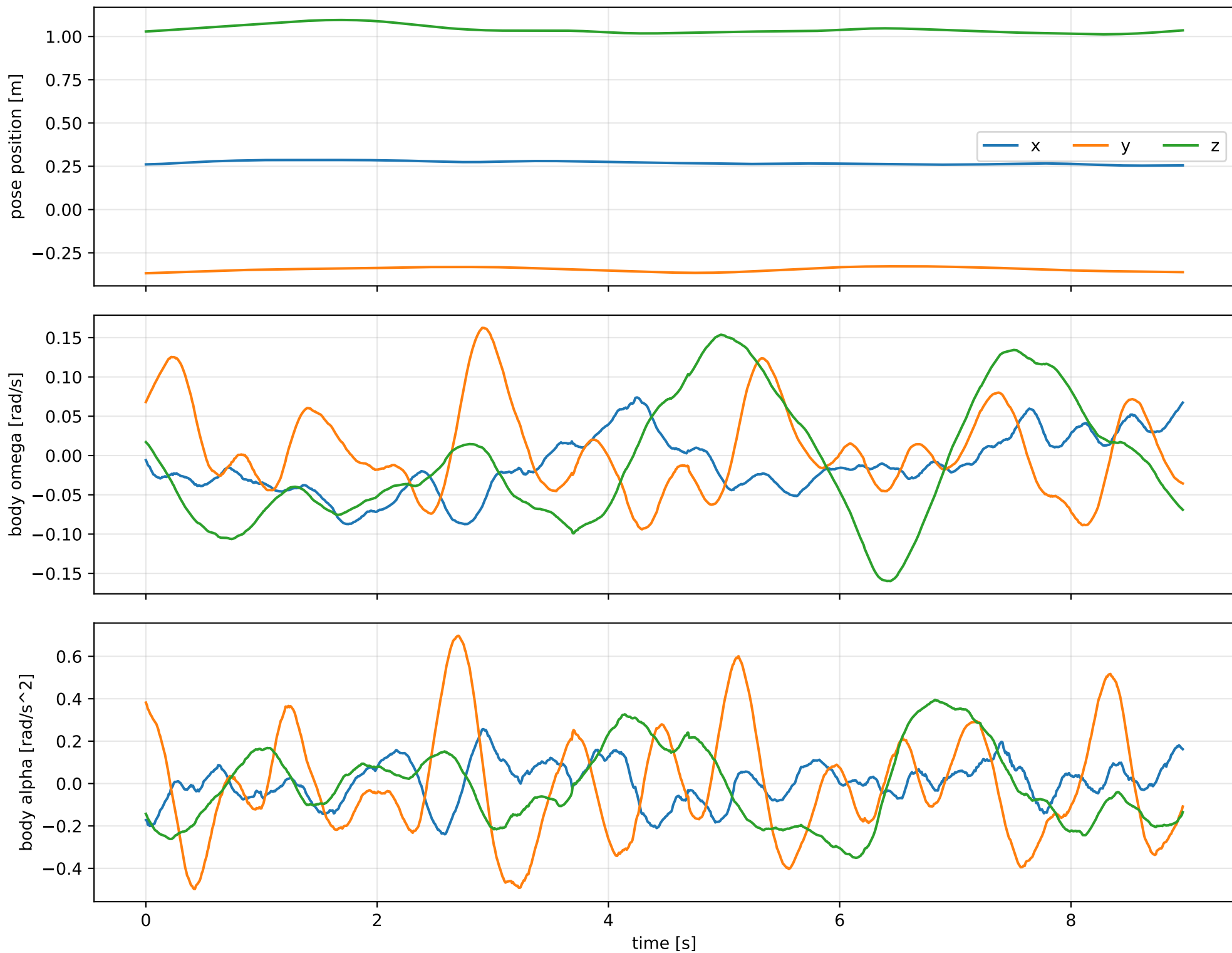
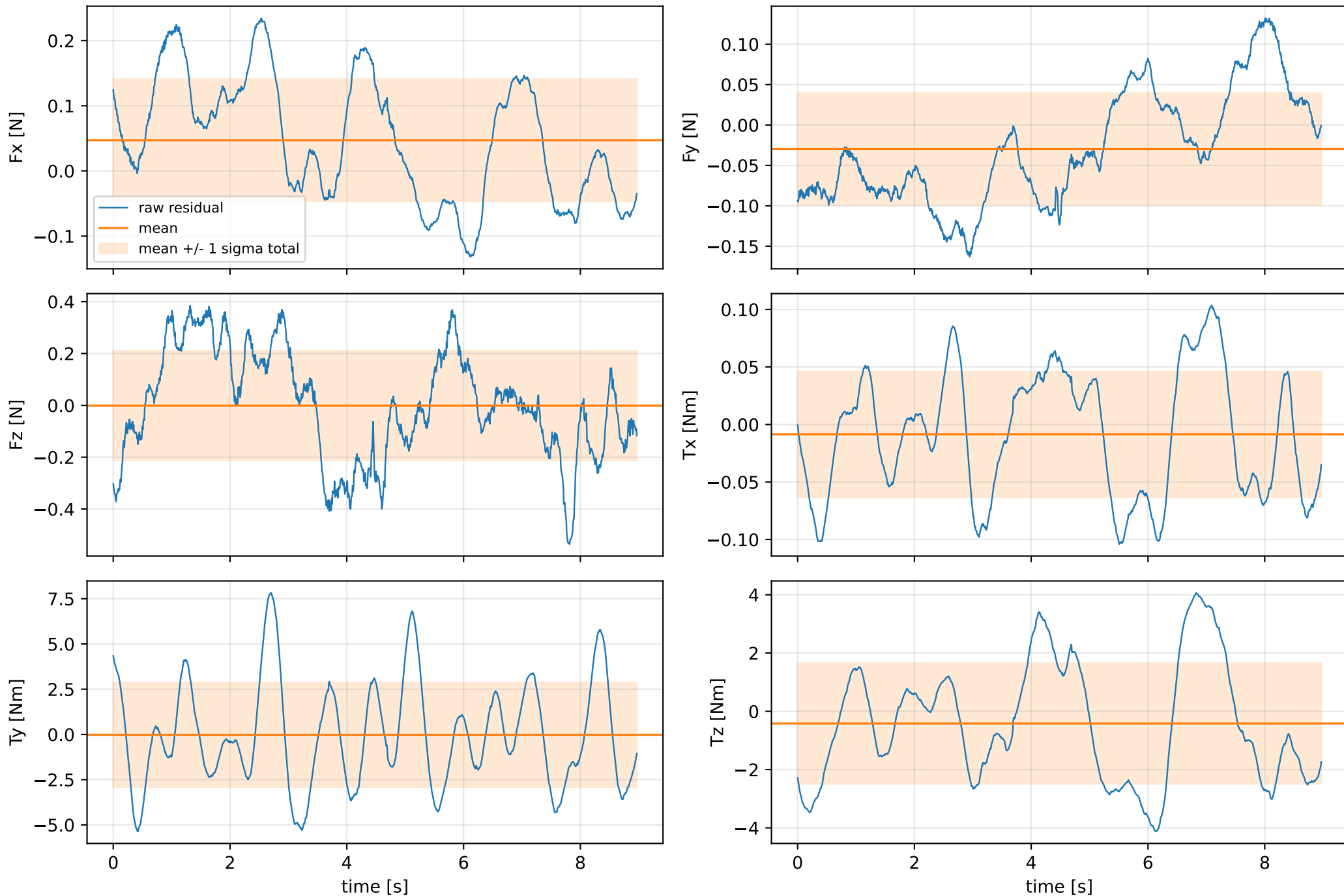


inertia_over_mass_xx_nominal: SG trajectory and derived motion



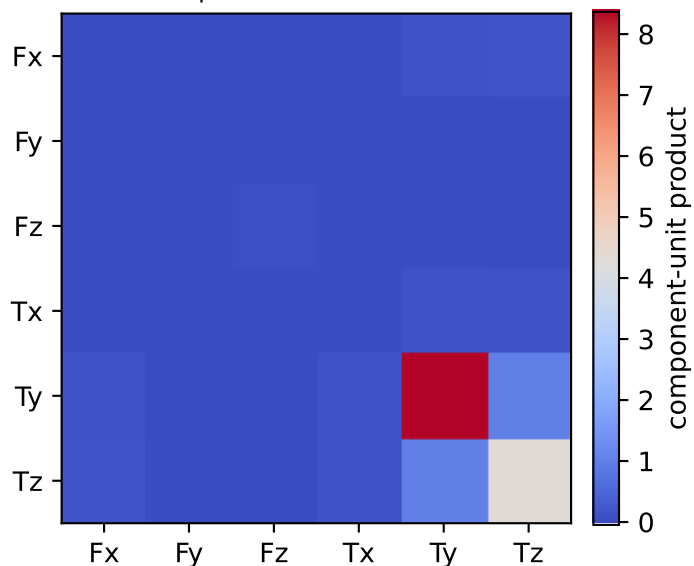
inertia_over_mass_xx_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



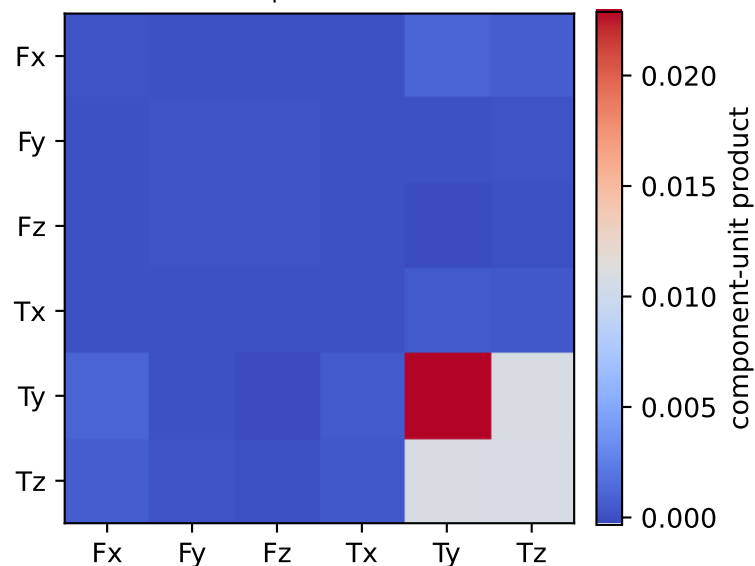
vector RMS about zero: force 0.2472 N, torque 3.5784 Nm; component std about mean: force [0.0936 0.0691 0.211], torque [0.0546 2.8921 2.068]

inertia_over_mass_xx_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [1.431e-05 1.306e-03 5.122e-03 4.562e-02 4.045e+00 8.571e+00]

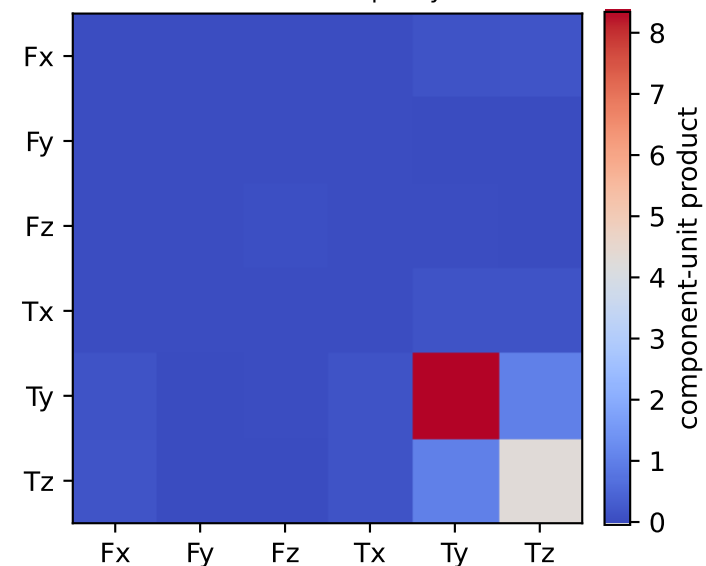
empirical total covariance



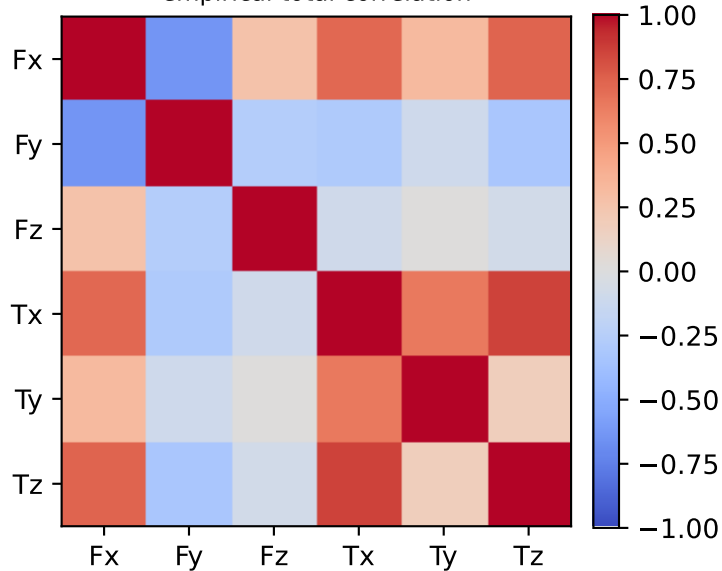
mean full-SG prediction covariance



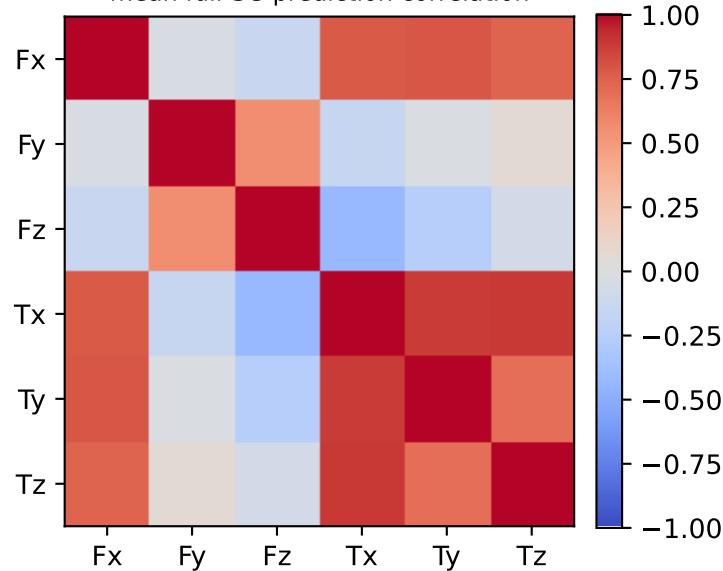
PSD excess model discrepancy covariance



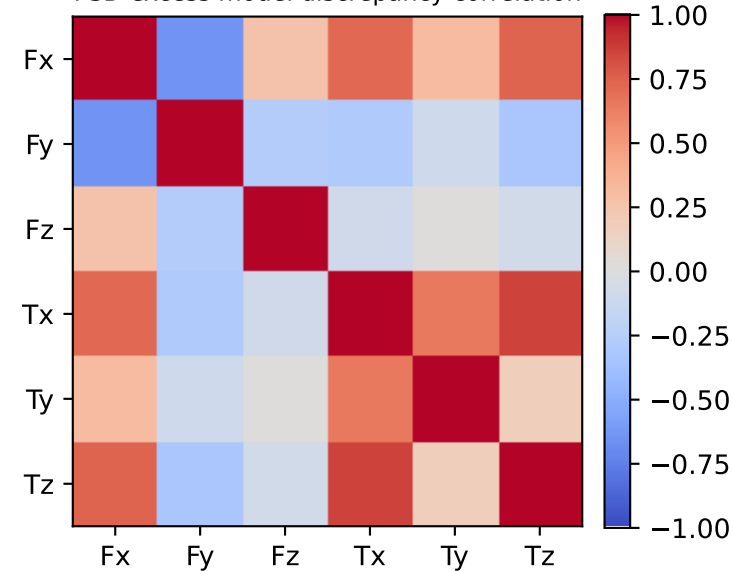
empirical total correlation



mean full-SG prediction correlation

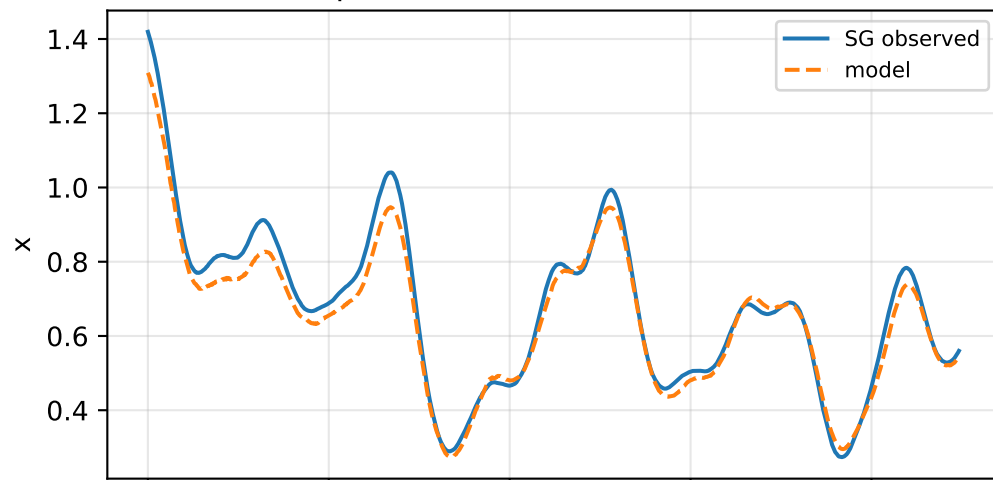


PSD excess model discrepancy correlation

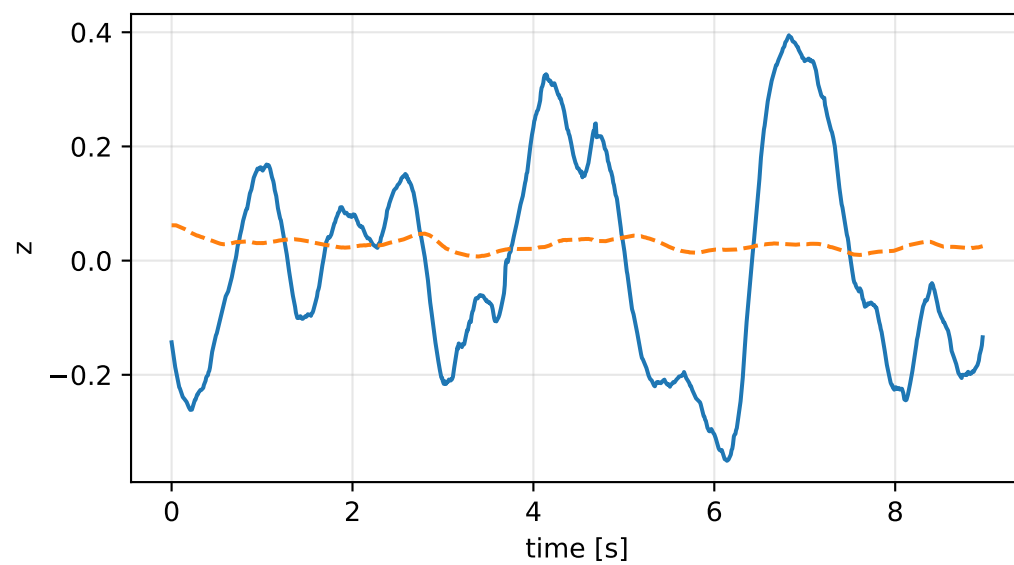
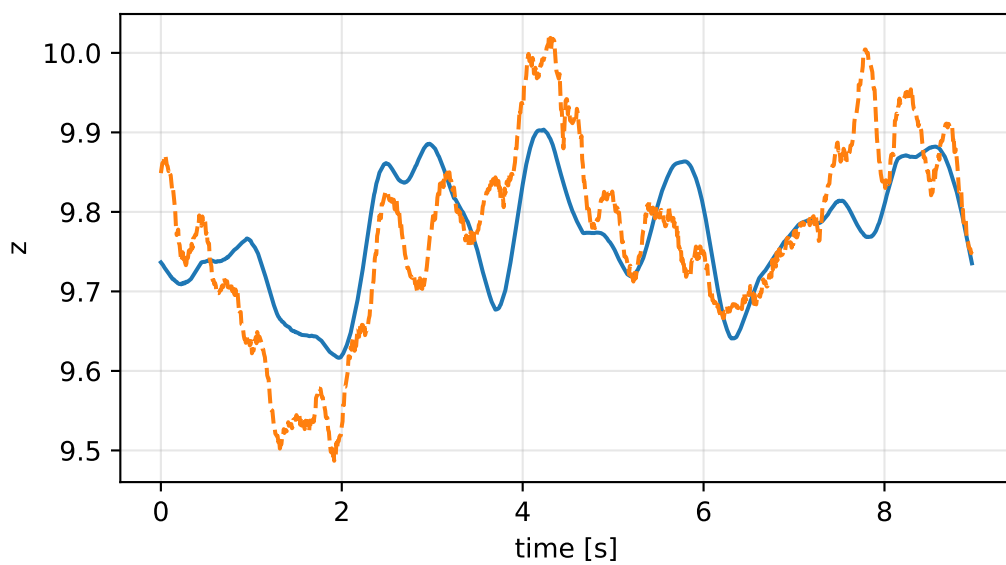
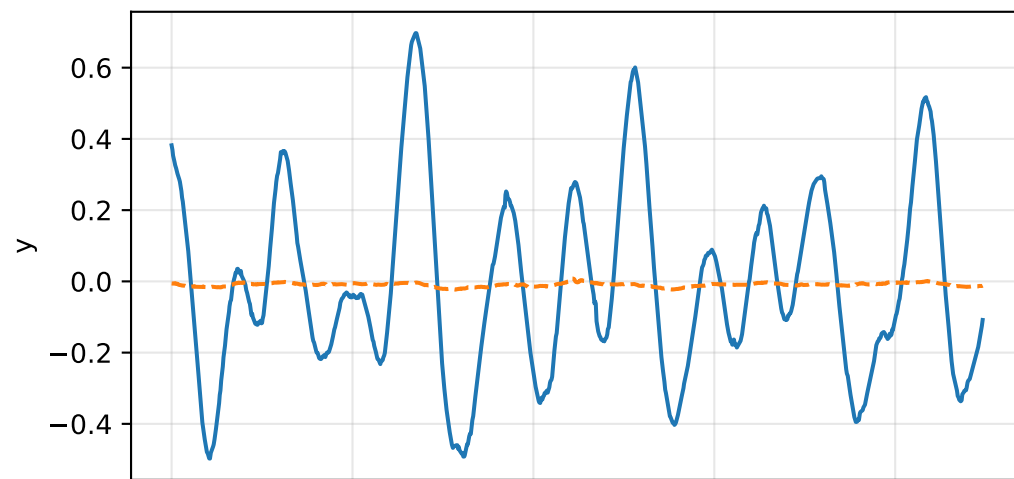
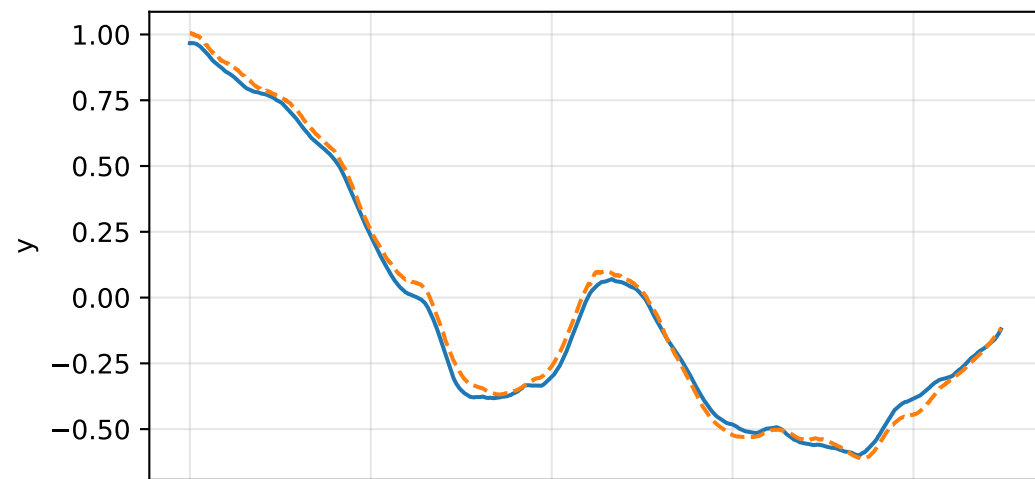
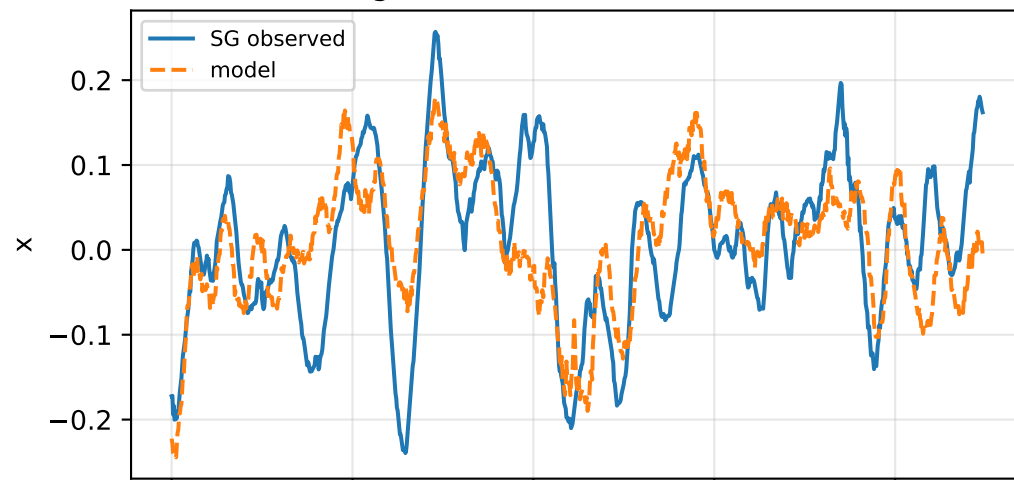


inertia_over_mass_xx_nominal: acceleration residual objective

specific acceleration [m/s^2]

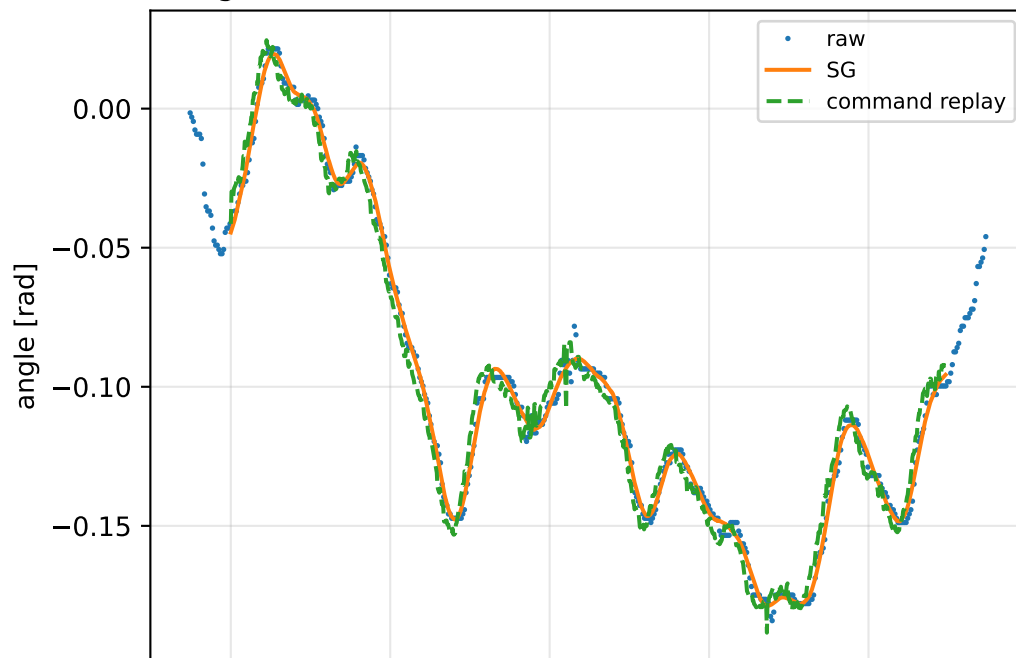


angular acceleration [rad/s^2]

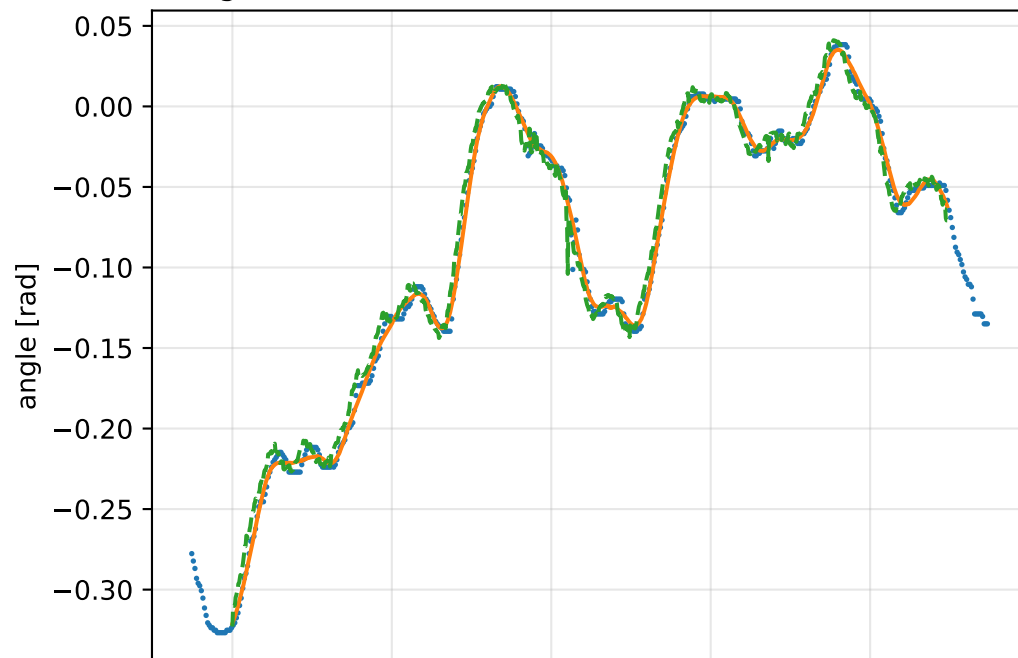


inertia_over_mass_xx_nominal: gimbal measurement audit (objective=measured_sg)

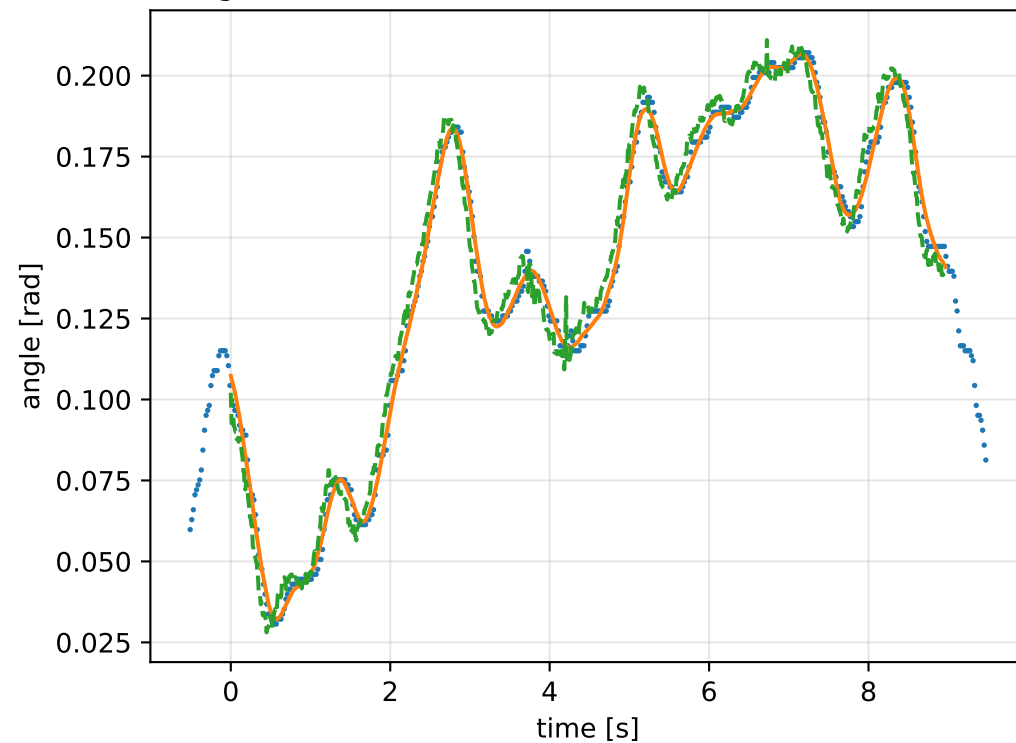
gimbal 1: RMSE=0.006345, max=0.01494 rad



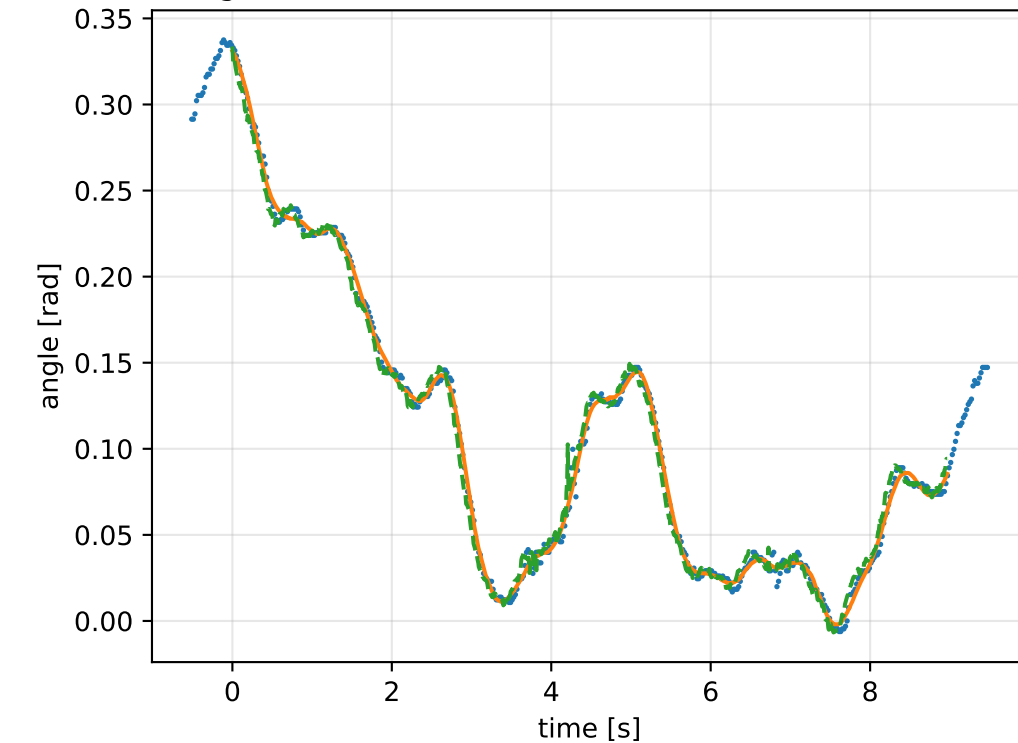
gimbal 2: RMSE=0.008835, max=0.04462 rad



gimbal 3: RMSE=0.007019, max=0.01611 rad

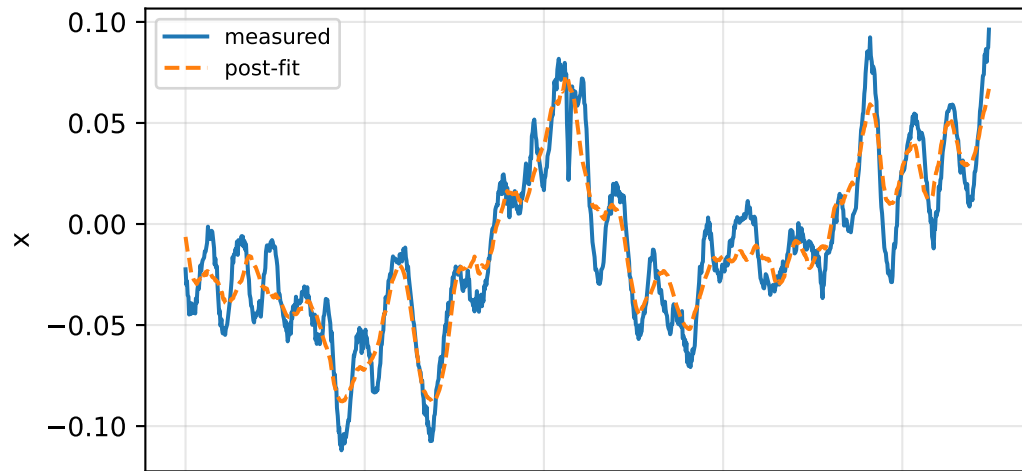


gimbal 4: RMSE=0.006321, max=0.03642 rad

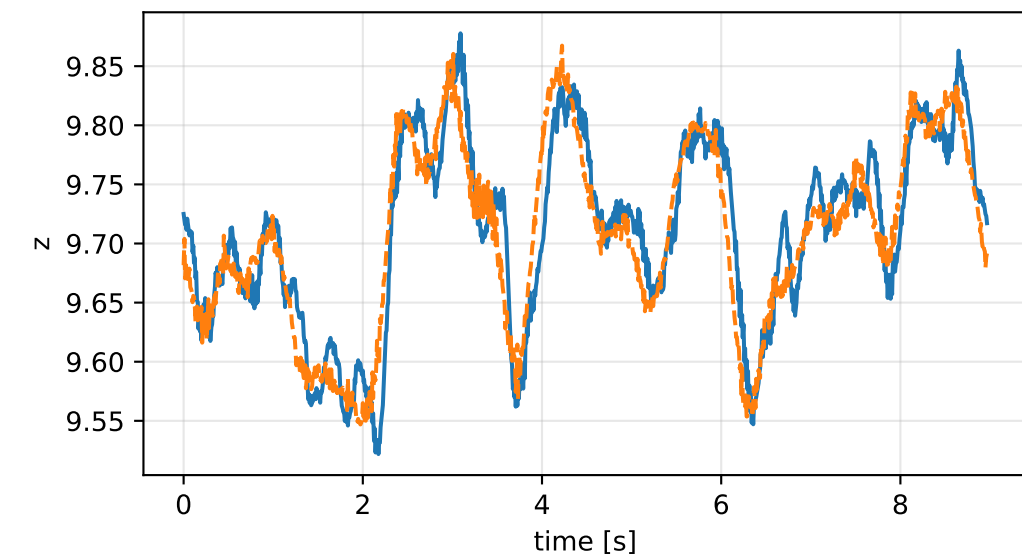
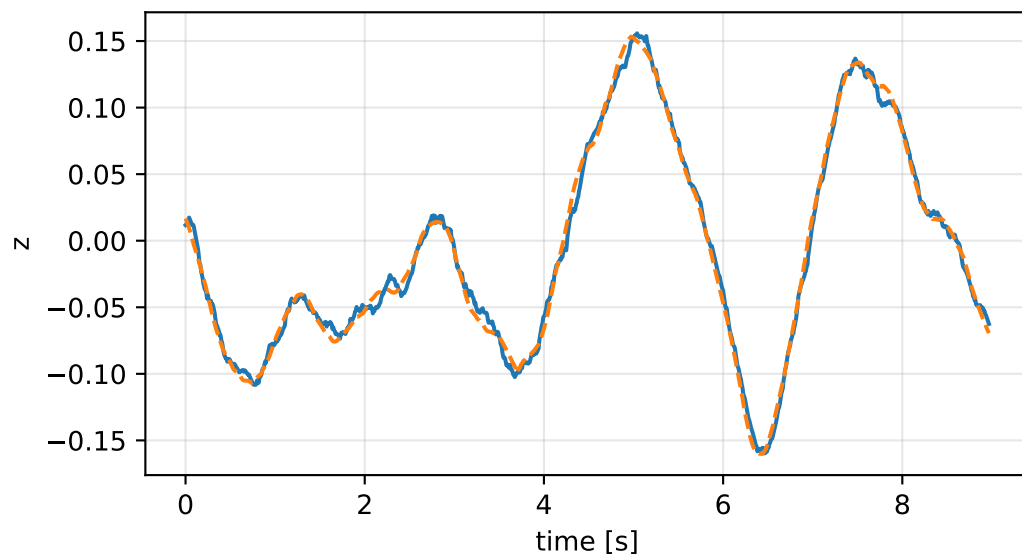
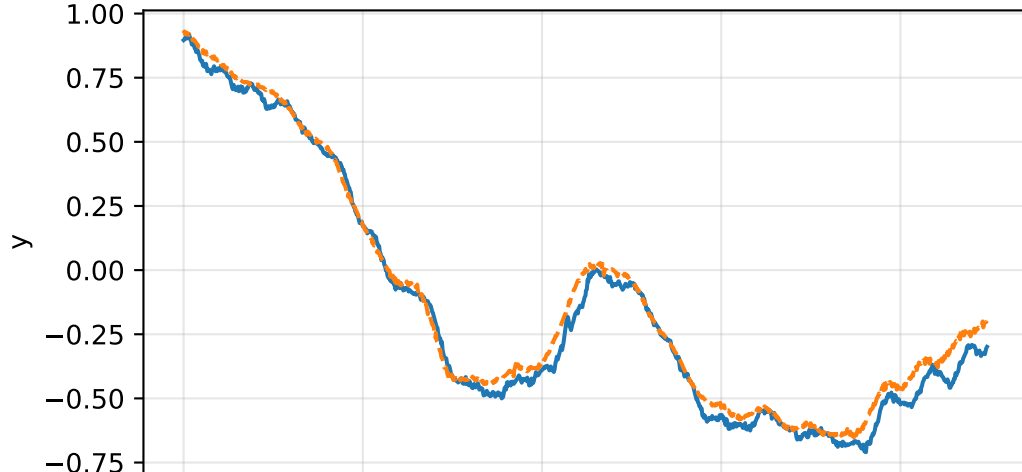
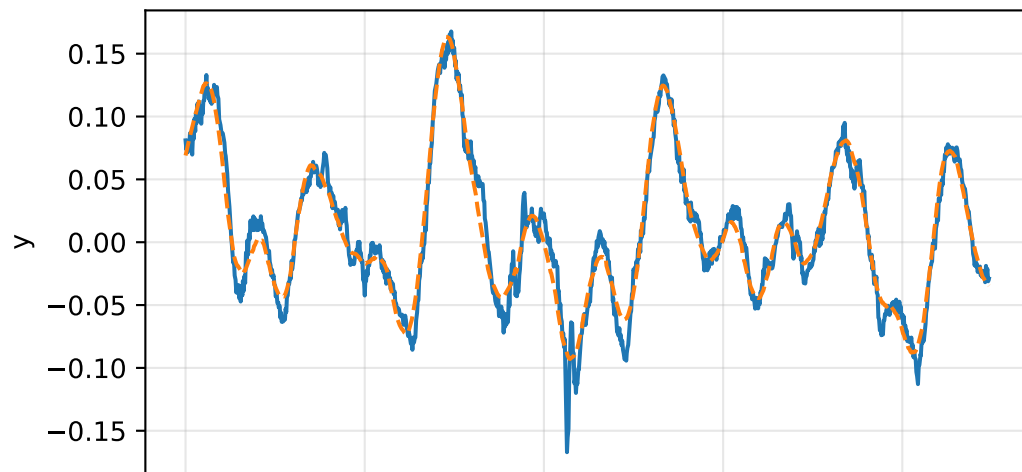
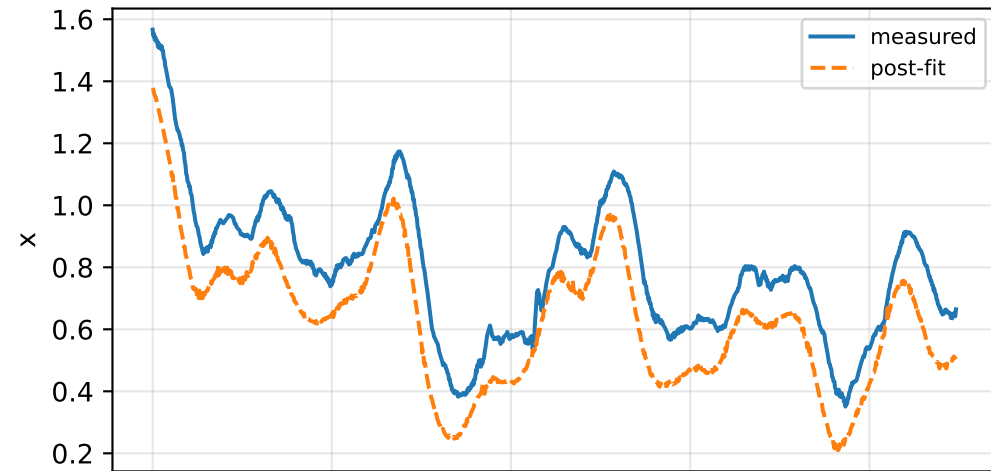


inertia_over_mass_xx_nominal: IMU comparison (diagnostic only)

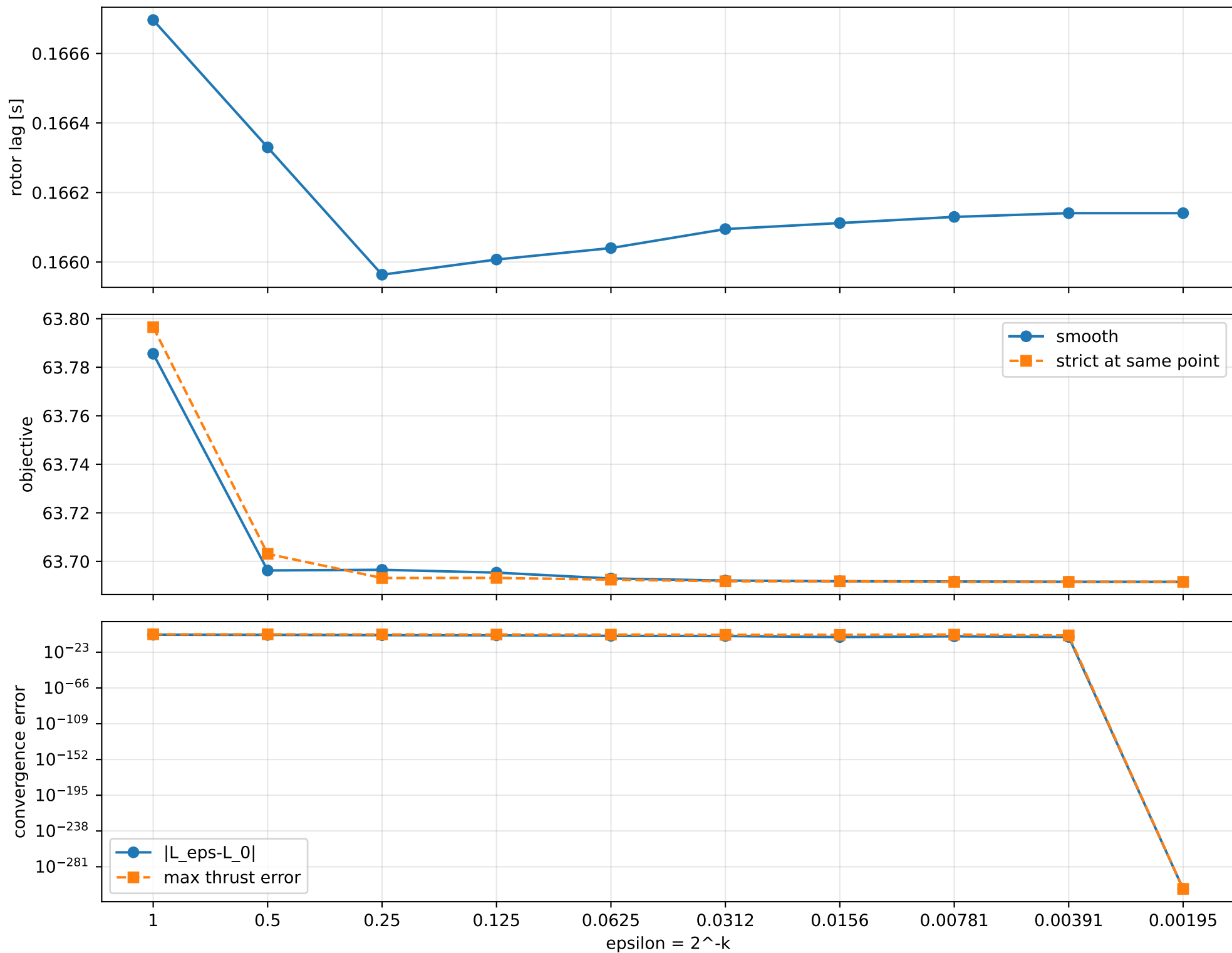
gyro [rad/s]



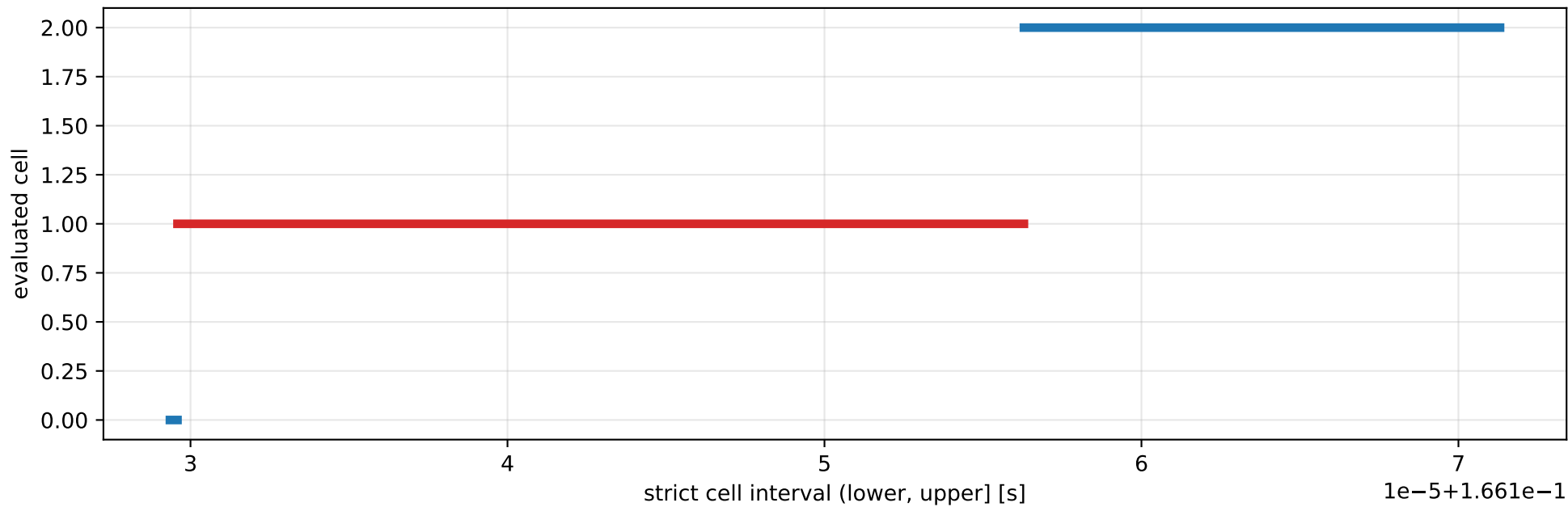
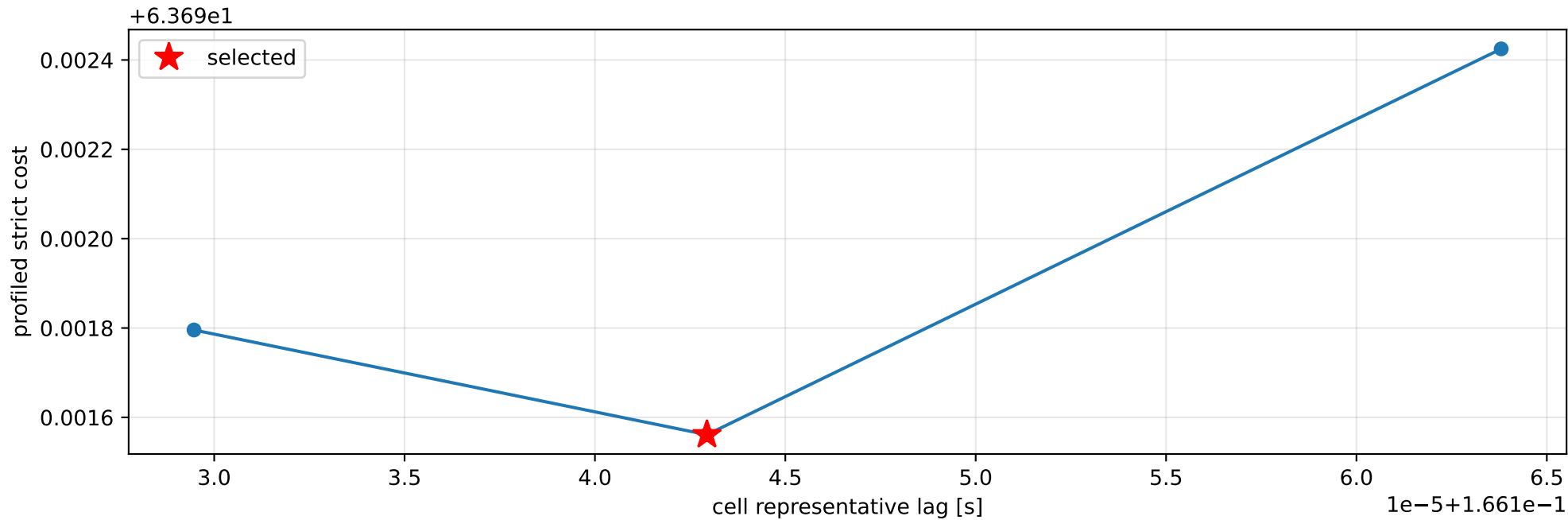
specific force [m/s^2]



inertia_over_mass_xx_nominal: smooth-to-strict continuation



inertia_over_mass_xx_nominal: exact strict-ZOH lag cells



selected (0.166129589, 0.166156292] s; final smooth 0.166140632 s; data boundary=False

inertia_over_mass_xx_nominal: parameter estimate

Case: inertia_over_mass_xx_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 2.35541741

inertia [kg m²]:

[[0.06510564 0.10825581 0.22376078]

[0.10825581 11.19572068 -0.02602366]

[0.22376078 -0.02602366 11.15824898]]

CoG body [m]: [-0.00170762 -0.20954163 0.01899749]

force effectiveness: [1.52410038 1.70693097 0.0455412 0.05292826]

scale-free J/m [m²]:

[[0.02764081 0.04596035 0.09499836]

[0.04596035 4.75317904 -0.01104843]

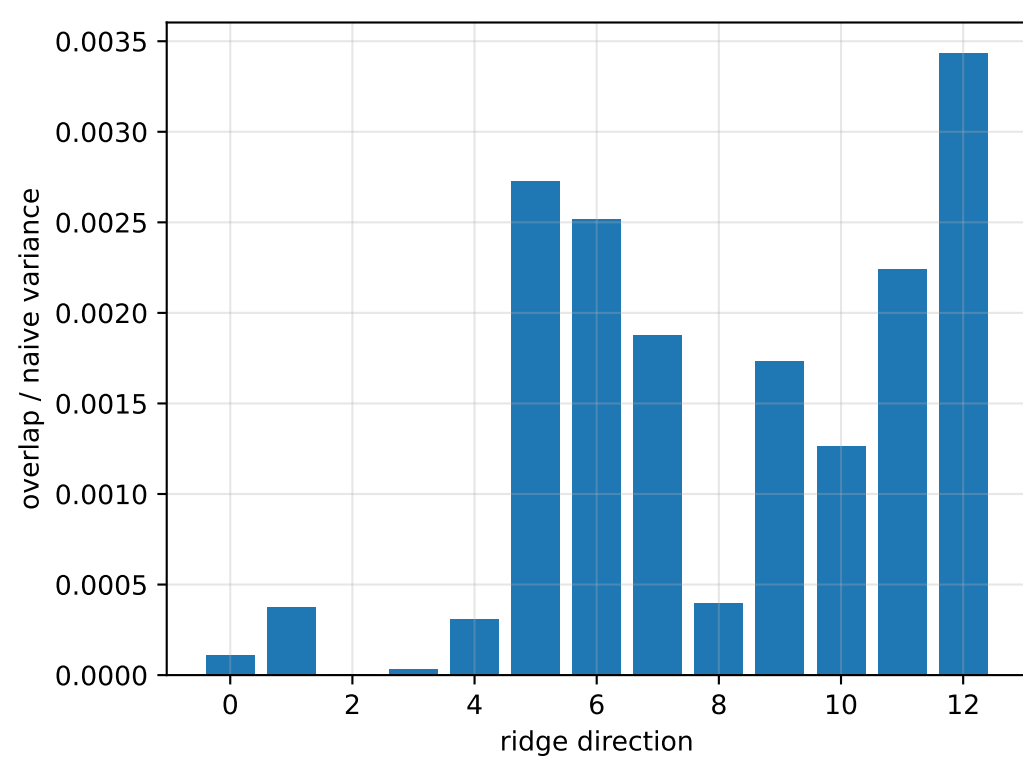
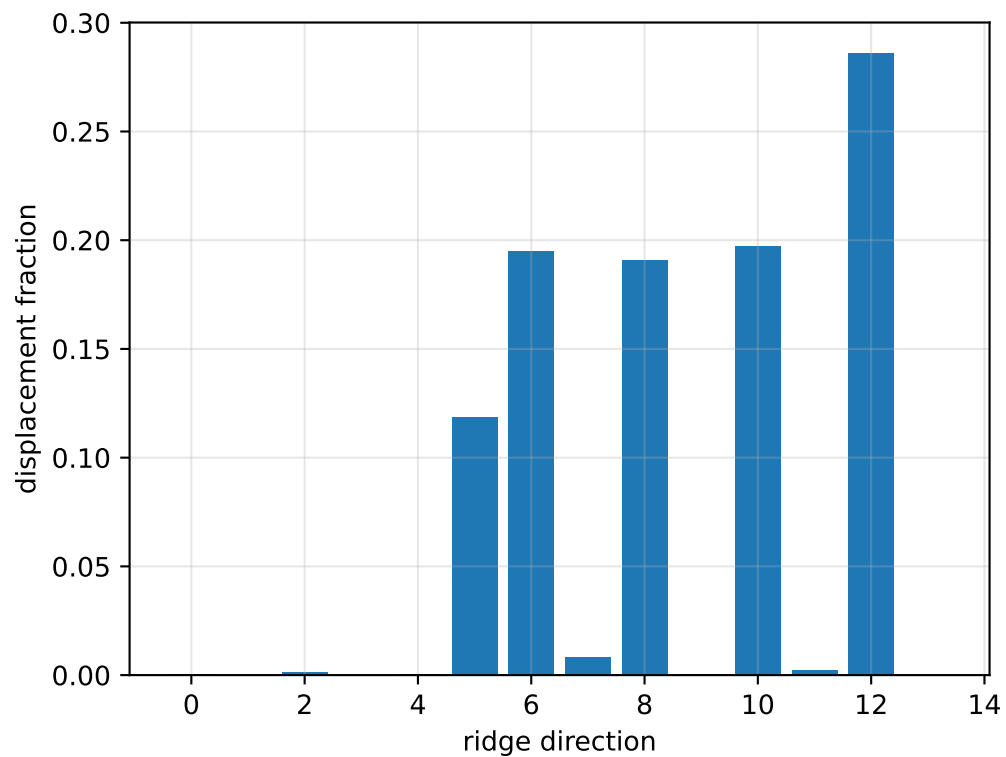
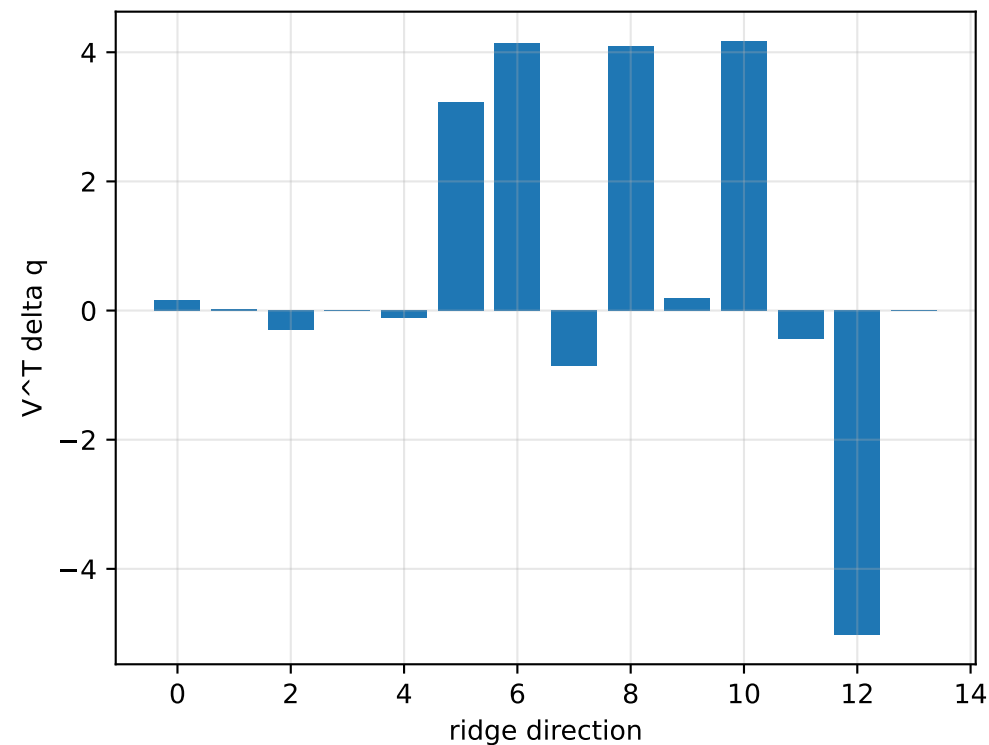
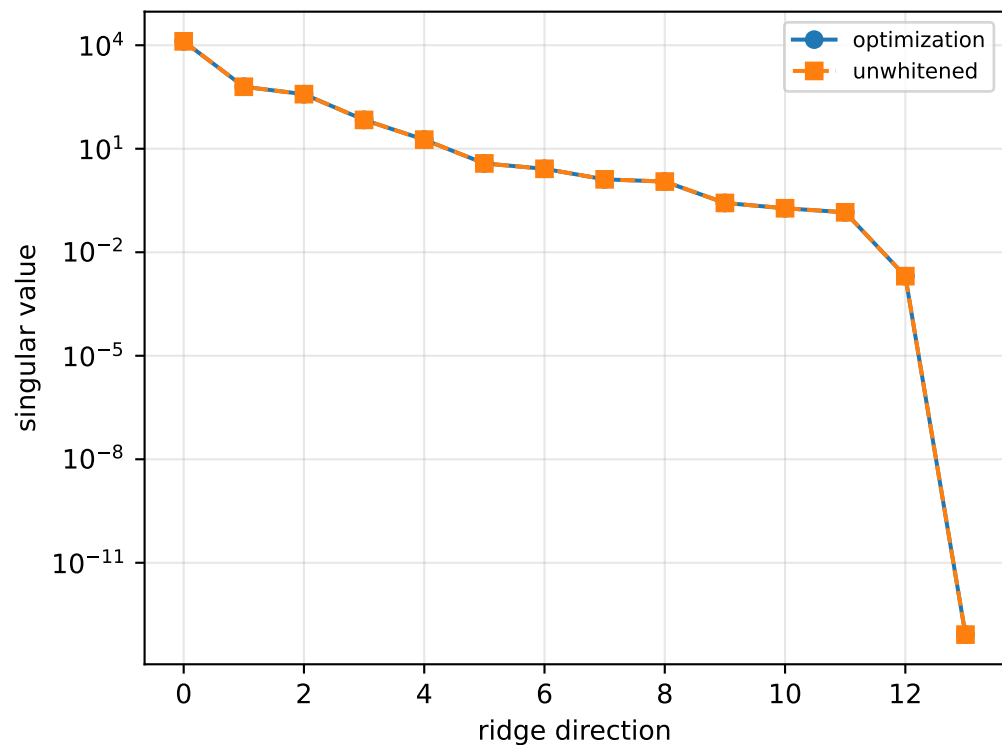
[0.09499836 -0.01104843 4.73727031]]

scale-free f/m: [0.64706169 0.72468301 0.01933466 0.02247086]

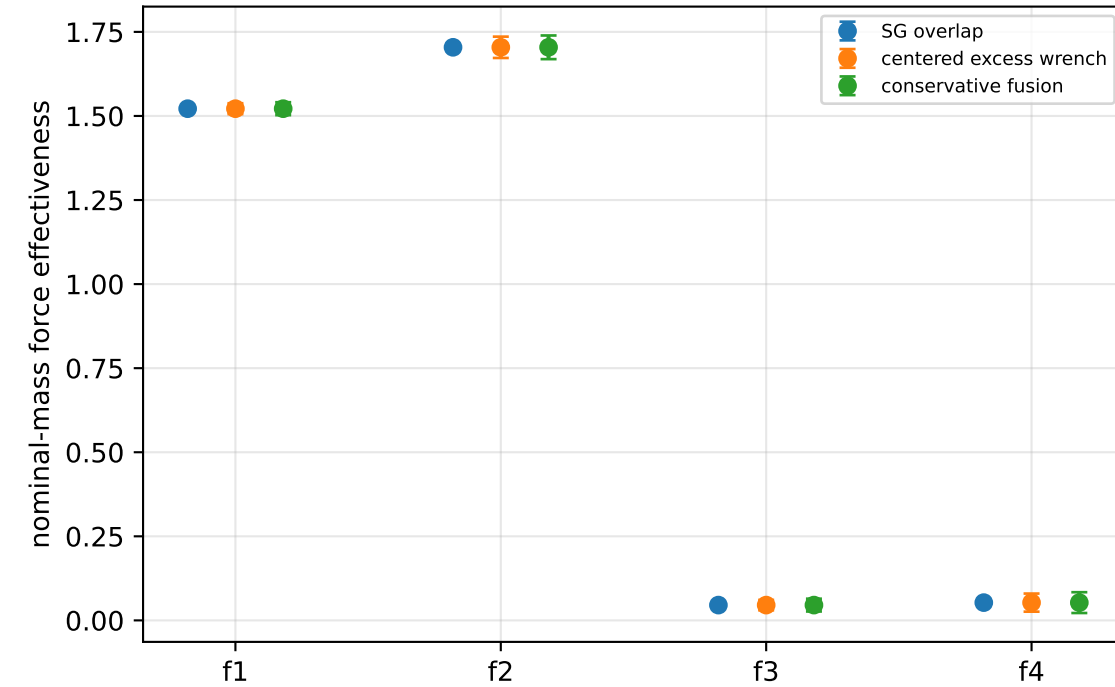
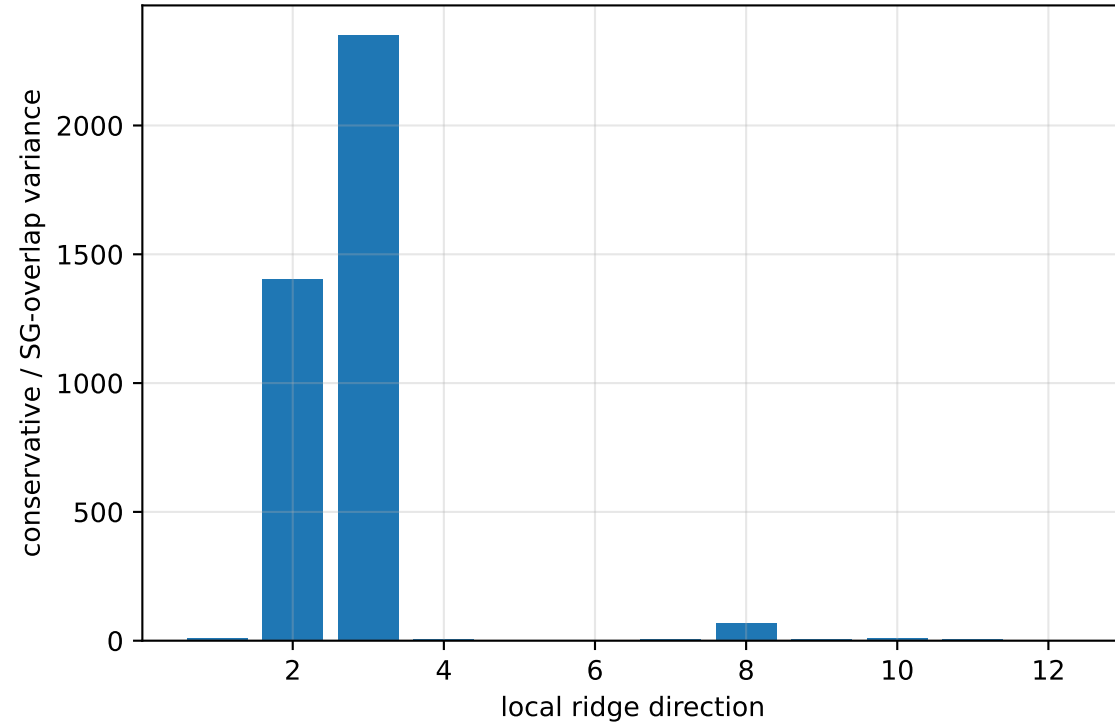
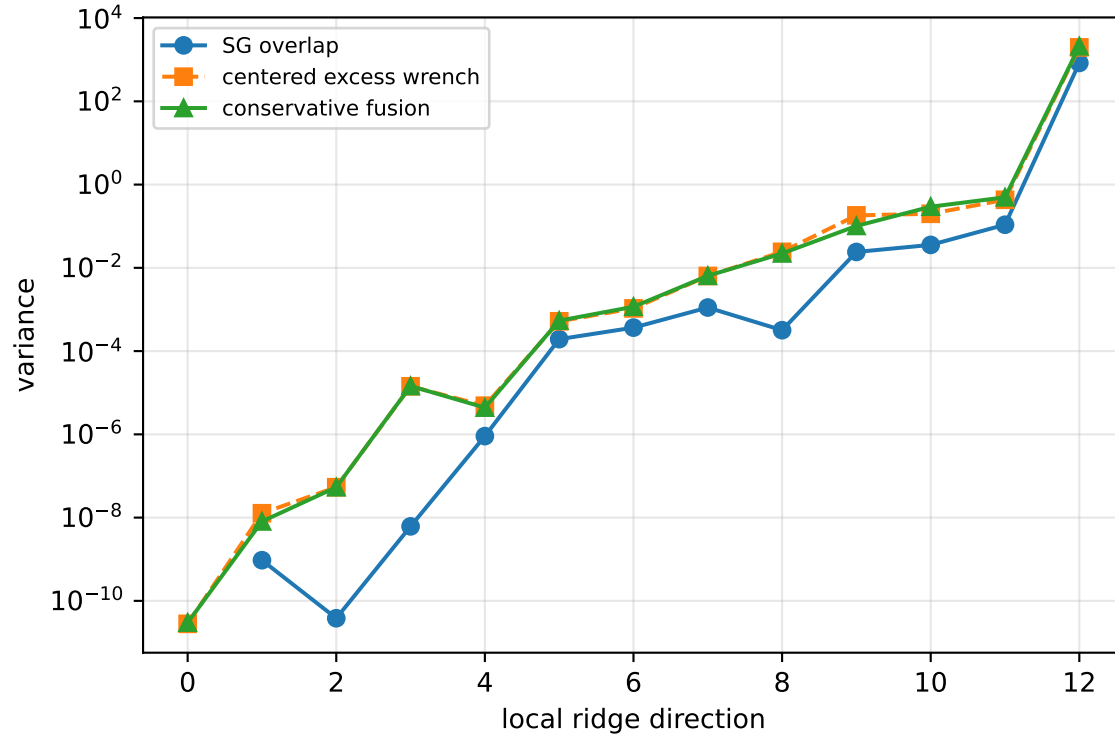
rotor lag cell: (0.166129589, 0.166156292] s

representative: 0.166142941 s

inertia_over_mass_xx_nominal: ridge spectrum (rank 13, nullity 1, ||jv||=6.28e-14)



inertia_over_mass_xx_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 44.24$
wrench-to-acceleration recovery max error = $3\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

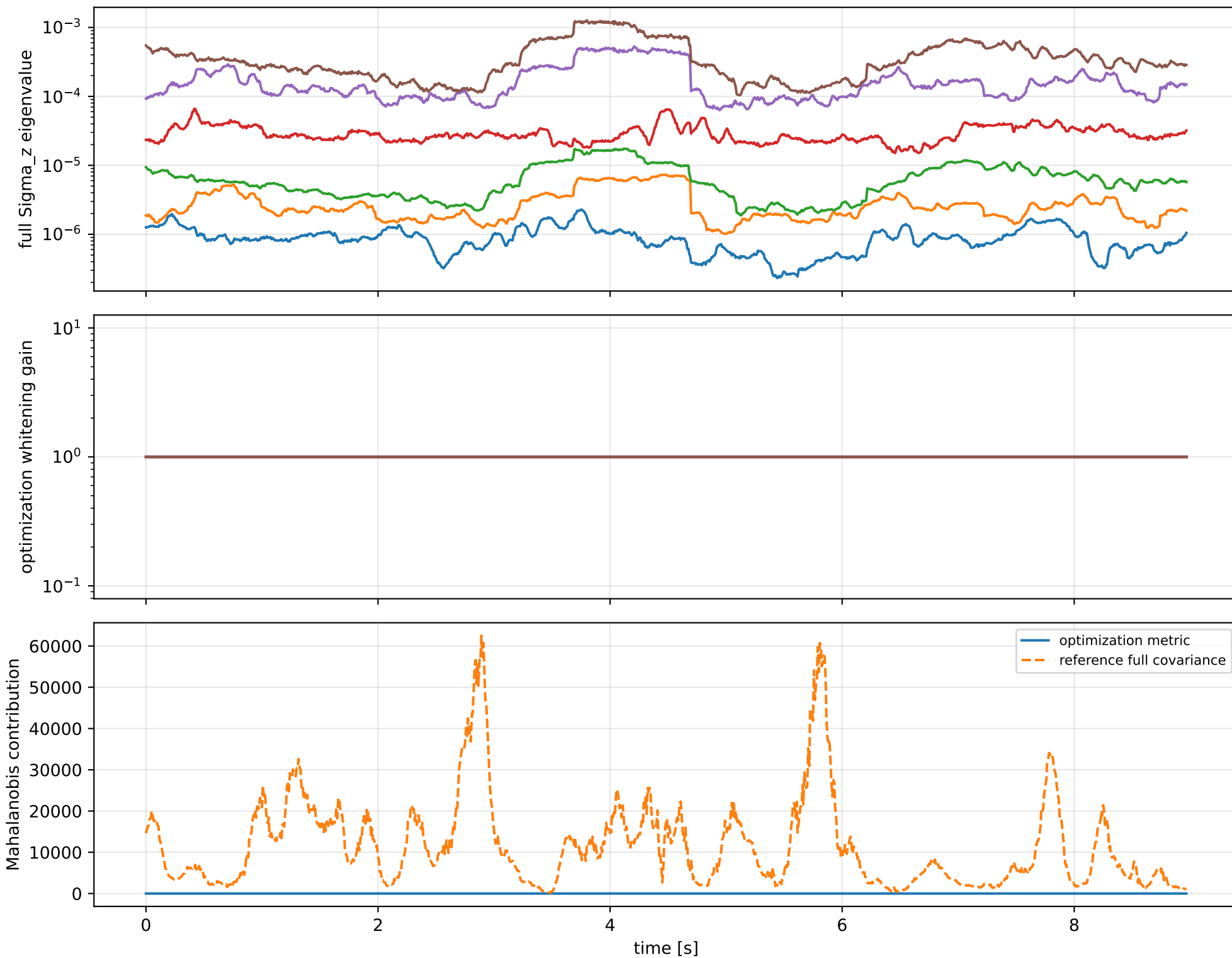
index 12: variance=2099, ratio=2.509
 $\log_{\text{mass}}=-0.133$; second_moment_diag_1=-0.133
second_moment_diag_2=+0.933; second_moment_diag_3=-0.138
second_moment_offdiag_12=-0.00173; second_moment_offdiag_13=-0.000921
second_moment_offdiag_23=-0.0711; cog_x=+7.04e-06
cog_y=+2.24e-06; cog_z=-9.24e-07
log_force_eff_1=-0.133; log_force_eff_2=-0.133
log_force_eff_3=-0.132; log_force_eff_4=-0.133

index 11: variance=0.4942, ratio=4.554
 $\log_{\text{mass}}=+0.198$; second_moment_diag_1=+0.256
second_moment_diag_2=+0.035; second_moment_diag_3=-0.634
second_moment_offdiag_12=+0.176; second_moment_offdiag_13=+0.006
second_moment_offdiag_23=+0.571; cog_x=+0.00122
cog_y=-0.00467; cog_z=-0.000309
log_force_eff_1=+0.206; log_force_eff_2=+0.212
log_force_eff_3=-0.0642; log_force_eff_4=-0.209

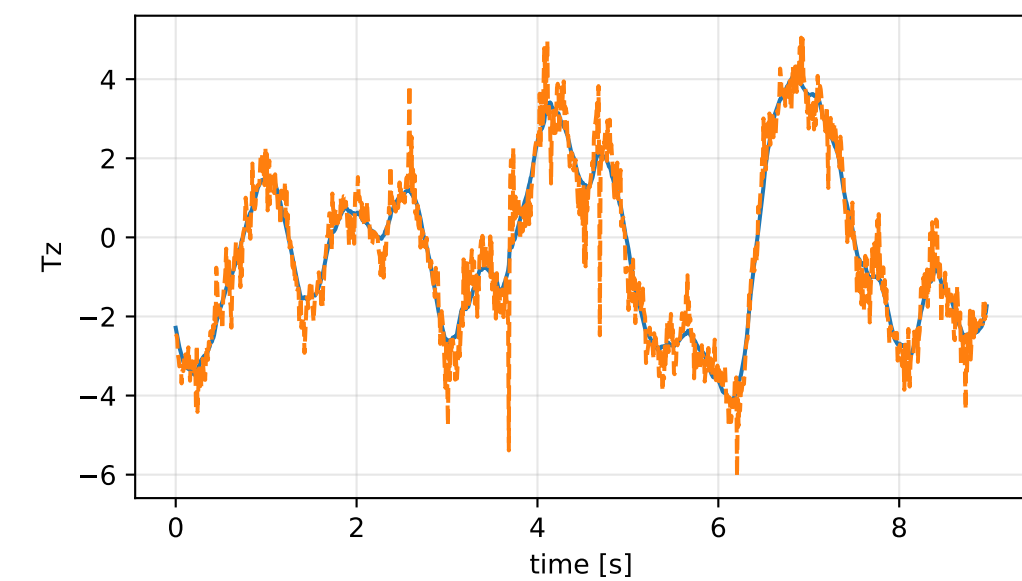
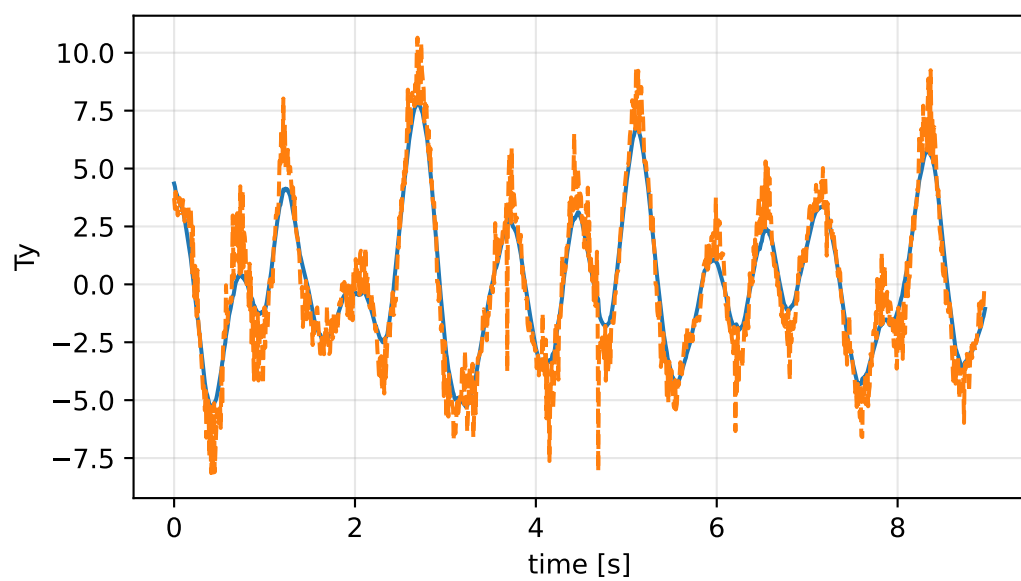
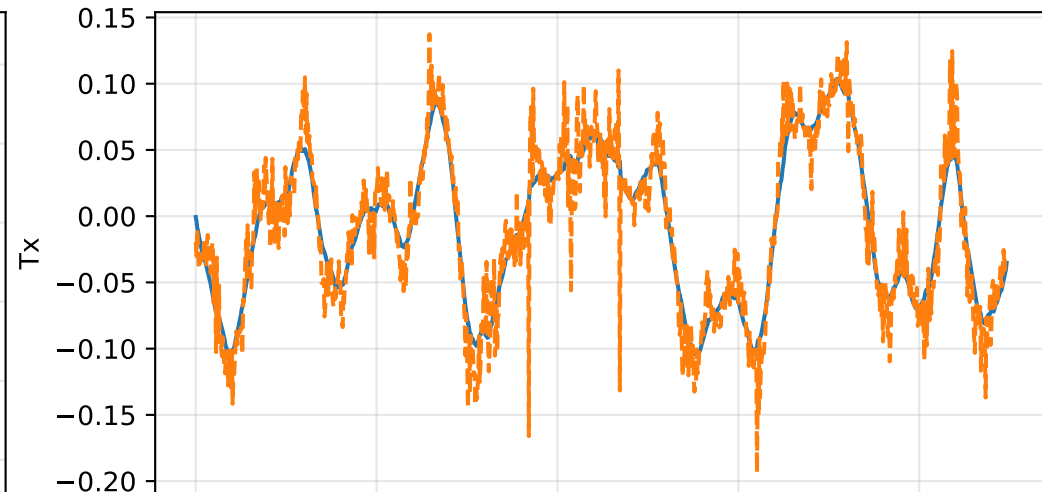
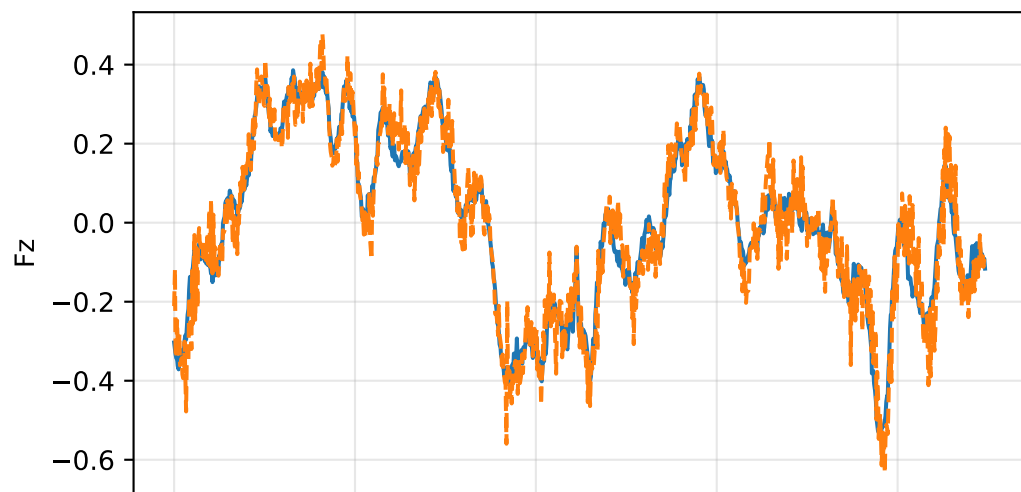
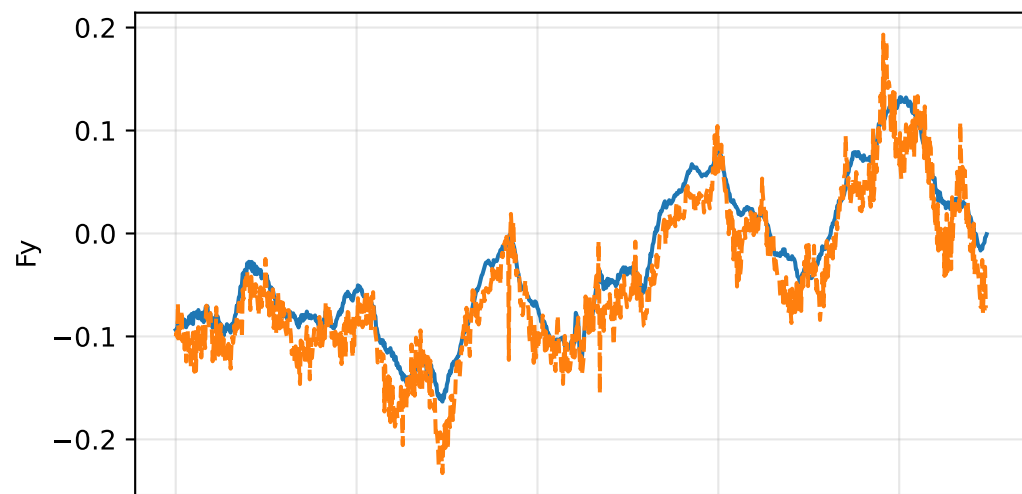
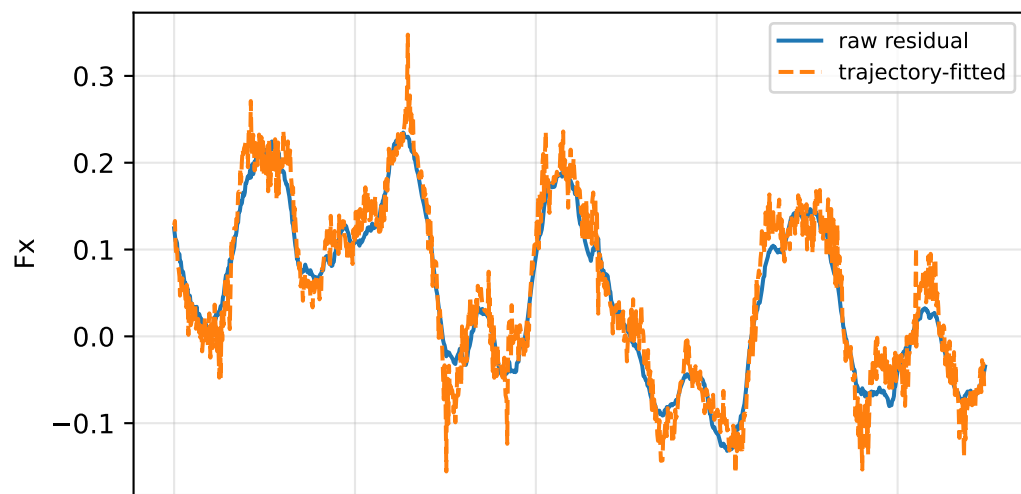
index 10: variance=0.294, ratio=8.265
 $\log_{\text{mass}}=+0.191$; second_moment_diag_1=+0.33
second_moment_diag_2=-0.0368; second_moment_diag_3=-0.0304
second_moment_offdiag_12=-0.157; second_moment_offdiag_13=+0.21
second_moment_offdiag_23=-0.553; cog_x=-0.000353
cog_y=-0.00845; cog_z=-0.000499
log_force_eff_1=+0.212; log_force_eff_2=+0.211
log_force_eff_3=-0.483; log_force_eff_4=-0.394

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

inertia_over_mass_xx_nominal: optimization vs reference covariance



inertia_over_mass_xx_nominal: wrench / closure (force max 7.17e-15, torque max 1.78e-15)



inertia_over_mass_xx_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_inertia_over_mass_xx_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/inertia_over_ma

SHA256: d50f14634ccbffa93995c113d524958bbbedaa410a60c8ea864b4c5cd041abdb4

data objective: 63.6915611703

prior objective: 2.72953689497e-10

total objective: 63.6915611706

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: inertia_over_mass_xx_nominal (inertia_over_mass_m2)

components: xx

target: [0.02764081]

std: [1.e-06]

final: [0.02764081]

error: [2.33646609e-11]

standardized residual: [2.33646609e-05]

factor objective: 2.72953689497e-10